

Self-Dual BKZ on NTRU: Dimension-Onset, Fatigue, and a Cross-Engine Reference-Free DSD Observable

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Abstract

This technical report extends the self-dual BKZ (SD-BKZ) reduction benchmark of the companion paper from *LWE-Kannan* lattices to *NTRU*, and from a single enumeration engine (fplll) to a second, sieving engine (G6K). We report three empirical results. First, on circulant NTRU lattices the SD-BKZ advantage exhibits a sharp *dimension-onset*: tied with BKZ for small n , a high-variance spike around $n \approx 71\text{--}73$ (mean advantage up to $+9.7$ nats), then a tight $+1.5$ -nat plateau for $n \geq 79$. Second, the defensible NTRU observable is *reference-free*: signed $d(\text{LN})$ is a reference artifact on NTRU (no canonical fixed point), but the BKZ-vs-SD-BKZ profile divergence / dense-sublattice-discovery (DSD) onset is robust, and the SD-vs-BKZ DSD-onset gap grows with dimension ($0\% \rightarrow 27\%$ over $n = 67\text{--}113$) as q crosses the fatigue point. Third, the cross-engine study confirms the companion paper’s thesis from the other side—the Root Hermite Factor is blind to the SD-BKZ difference while $d(\text{LN})/\text{DSD}$ sees it—now on a sieve oracle, and finds the effect to be regime-dependent. We also re-validate, on a new modulus family, an fplll GSO numerical fix. All experiments are reproducible via a containerised, determinism-gated benchmark.

Keywords: lattice reduction, BKZ, SD-BKZ, NTRU, overstretched, fatigue, dense sublattice discovery, Rankin profile, sieving, G6K

1 Introduction

The companion paper [?] showed, on *LWE-Kannan* lattices, that the Root Hermite Factor (RHF) is structurally blind to differences between BKZ and self-dual BKZ [?] that the Li–Nguyen Rankin profile distance $d(\text{LN})$ [?] makes visible. This report asks two follow-up questions a data paper is well suited to answer: (i) *does the SD-BKZ phenomenology generalise beyond LWE to NTRU?* and (ii) *does it reproduce on a fundamentally different reduction oracle—sieving (G6K) rather than enumeration?*

Our contributions are primarily empirical and infrastructural:

- A characterisation of the SD-BKZ advantage on circulant NTRU lattices [?], including a sharp dimension-onset transition (Section ??).
- Evidence that the appropriate NTRU observable is reference-free, and a *dense-sublattice-discovery (DSD) onset* whose SD-vs-BKZ gap grows with dimension as q approaches the fatigue point (Section ??).

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Table 1: SD-BKZ advantage on NTRU: tied $\rightarrow$ onset spike $\rightarrow$ plateau.

n	mean adv	std	seeds > 0
59–67	~ 0	tight	3–9/20
71	+4.35	5.59	16/20
73	+9.68	5.21	19/20
79–89	+1.5	~ 0.2	20/20

- A cross-engine study (Section ??): G6K wired as a second engine behind a reproducibility-gated seam, a self-dual pump-and-jump construction validated against fplll’s SD-BKZ, and confirmation that RHF-blindness is engine-independent but regime-dependent—consistent with independent observations of RHF-equality under a non-exact oracle [?].
- An independent re-validation of an fplll GSO catastrophic-cancellation fix on a new (NTRU) modulus family (Section ??).

This is a *technical report / data paper*: the emphasis is on the experimental apparatus, the data, and reproducibility, rather than on new theory. Closest prior work is Deng and Jia’s self-dual BKZ over a sieve oracle [?]; we differ in measuring the SD-vs-BKZ structural difference via the reference-free profile observable rather than swapping the oracle for speed. White [?] studies dynamic block-size strategies within the same Li–Nguyen framework, complementary to the fixed- β , q -swept regime characterised here.

2 Background

2.1 Circulant NTRU lattices and fatigue

2.2 BKZ, SD-BKZ, and the two engines

2.3 The reference-free DSD observable

3 Experimental Setup

3.1 Generators and engines

3.2 Reproducibility and determinism gates

3.3 Campaigns and parameter grids

4 The SD-BKZ Advantage Has a Dimension-Onset on NTRU

On circulant NTRU at $q = 97$ (standard regime), $\beta = 20$, 50 tours, the SD-BKZ-vs-BKZ advantage has a *sharp* dimension transition, not a smooth ramp (20 seeds per n):

Three regimes: tied for $n \leq 67$; an onset zone at $n = 71$ –73 where the mean spikes to +4–10 nats with $\sim 25\times$ the plateau variance (SD-BKZ cracks the dense sublattice on most-but-not-all instances); and a stable +1.5-nat plateau for $n \geq 79$ (std ≈ 0.2 , 20/20 wins). The onset spike exceeds the plateau—non-monotonic.

Table 2: Reference-free DSD-onset: SD-BKZ reaches dense-sublattice discovery at lower q than BKZ, gap growing with n (~ 15 – 20 seeds/cell).

n	SD onset q	BKZ onset q	gap%
67	146	149	2
79	175	175	0
89	237	281	18
101	426	514	21
113	732	932	27

5 Fatigue and the Cross-Dimension DSD-Onset Gap

Sweeping q across the fatigue point at fixed n , the reference-free BKZ-vs-SD-BKZ profile divergence peaks near $q \approx q_{\text{fat}}$. The SD-BKZ DSD-onset occurs at progressively *lower* q than the BKZ onset as n grows—the gap widens with dimension:

$n = 127$ is excluded (fppll §8 catastrophic cancellation + an off-grid crack).

6 Cross-Engine: SD-BKZ on a Sieve Oracle

RHF blindness is engine-independent but regime-dependent. On NTRU at $q = 97$, $\beta = 40$, the companion paper’s thesis reproduces on the sieve: in the standard regime the RHF of BKZ and SD-BKZ bases is identical to machine precision while $d(\text{LN})$ differs. The effect is regime-dependent—RHF-blindness breaks only on the rare seeds where SD-BKZ finds a short vector (which RHF reflects). This is consistent with RHF-equality reported under a non-exact (approximate-enumeration) oracle [?]; the present work measures the structure RHF misses.

Cross-engine validation of the construction. A dimension-resolved fppll-vs-G6K comparison ($n = 89/101/113$, 20 seeds each) confirms the self-dual G6K construction is sound: at $n = 101/113$ the two engines agree perfectly (no spurious sieve firing), and where a short vector genuinely exists ($n = 89$, seed 19) both engines land the identical first GS log-norm.

The gap-regime and the sieve-meaningful block size do not overlap. A G6K DSD-onset q -sweep at $n = 89$, $\beta = 40$, over $q \in \{211, 257, 307, 401, 521\}$ finds *both* SD-BKZ and BKZ at full dense-sublattice discovery on every seed at every q (short = $89 = n$; $b_1 \approx 1.96$, the q -independent ternary-secret floor $\sqrt{2n \cdot 2/3}$), with SD-vs-BKZ advantage ≈ 0 . The whole window is post-onset—no gap. This is informative: the DSD-onset *gap* (Table ??; SD reaching discovery at lower q than BKZ, the gap growing with n) was measured at $\beta = 20$, whereas the sieve oracle needs $\beta \geq 40$ to do meaningful work, at which point both engines crack DSD together. The gap-regime ($\beta = 20$) and the sieve-usable regime ($\beta \geq 40$) thus *do not overlap* at $n = 89$: the SD-vs-BKZ onset advantage is a weak-reduction (low- β) phenomenon that a strong sieve oracle erases by finding the dense sublattice on both passes.

7 Re-Validating the fppll GSO Fix on a New Modulus Family

The companion paper documents a catastrophic-cancellation bug in fppll’s squared-form GSO update at cryptographic moduli, and a Kahan-compensated fix. We re-validate

that fix on a new (NTRU) modulus family: on $q \in \{971, 1087, 1201\}$ at $n = 127$, the four contaminated seeds recover from the precision-floor clamp ($b_1 = -345.388$) into the control band (≈ -0.1); the patched build passes all 15 fpml regression tests. The fix is thus validated on two modulus families (the companion paper's $q = 3329$ LWE, $38\% \rightarrow 0\%$ degeneracy, and this NTRU set).

8 Discussion

9 Reproducibility

10 Conclusion